High-Symbol-Rate 192-GBaud Signal Transmission in S+C+L Band over 2000 km with Net Bitrate of 102.8 Tb/s

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Abstract: We demonstrated 92-channel 192-GBaud WDM transmission in 18.4-THz triple-band with our proposed low-noise forward Raman pumping technique, achieving the highest average channel rate of 1.117 Tb/s/λ over 2000 km with a total net-bitrate 102.8 Tb/s. © 2026 The Author(s)

1. Introduction

High-capacity long-haul optical transmission with high-speed digital coherent transceivers is essential for core networks to address the rapid increase in communication traffic cost-effectively. To achieve this, the transceiver capacity per wavelength (b/s/λ) should be increased to accommodate high-speed client signals such as next-generation Ethernet. 200-GBaud-class wavelength division multiplexed (WDM) transmission experiments with high-symbol-rate signal handling technologies have been reported [1–3], achieving more than 2-Tb/s/λ full C-band WDM transmission over 1040 km with a total net bitrate of 36.9 Tb/s [3]. To further increase the capacity to over 100 Tb/s in single-mode fiber (SMF) transmission while maintaining the transmission distance, multiple-band WDM transmission technologies are needed that have high-speed digital coherent signals [4]. As shown in Fig. 1, 100-Tb/s-class long-haul transmission experiments have been reported with multiple-band WDM technologies [4–15]. Recently, we demonstrated 105.6-Tb/s transmission over 2000 km using our proposed low-noise forward (FW) pumped distributed Raman amplification (DRA) with 122 channels of 144-GBaud signals with an average channel rate of 0.865 Tb/s/λ [10].

In this study, we applied high-symbol-rate 192-GBaud signals to the triple-band WDM transmission that utilizes our proposed low-noise FW-pumped DRA technique [10, 16, 17] to further increase the channel rate while maintaining the transmission distance. 100-Tb/s-class high-capacity long-haul transmission was achieved with a net bitrate of 129.3 Tb/s over 1040 km and 102.8 Tb/s over 2000 km with 92 channels of 192-GBaud probabilistically constellation-shaped (PCS) quadrature amplitude modulation (QAM) signals within a 200-GHz WDM grid under an 18.4-THz S+C+L-band WDM configuration. As shown in Fig. 1, we successfully demonstrated that the high-symbol-rate 192-GBaud signals enabled, to the best of our knowledge, the highest average channel rate of 1.406 Tb/s/λ at 1040 km and 1.117 Tb/s/λ at 2000 km for 100-Tb/s-class SMF inline-amplified transmission.

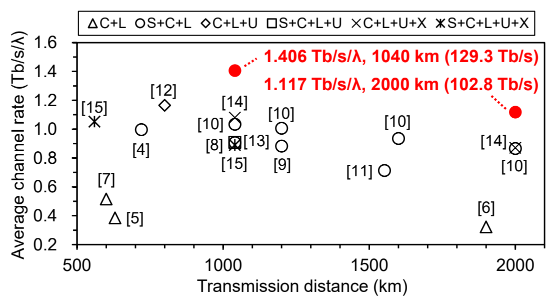

Fig. 1. Average channel rate vs. transmission distance for 100-Tb/s-class SMF inline-amplified transmission.

2. Experimental Setup

We conducted the high-capacity long-haul transmission experiments using the high-symbol-rate 192-GBaud signal in the S+C+L band with the setup in Fig. 2(a). On the transmitter (Tx) side, the main signal with the symbol rate of 192 GBaud was generated by using an offline Tx digital signal processing (DSP), a 256-GSa/s 80-GHz arbitrary waveform generator (AWG), a thin-film lithium niobate (TFLN) IQ modulator (IQM) with a 5-dB bandwidth of 110 GHz, and a polarization division multiplexing (PDM) emulator. The offline Tx-DSP has functional blocks including PCS-QAM symbol generation, pilot quadrature phase shift keying symbol insertion of every 64-symbol duration with a pilot overhead of 1.5873% (= 100 × (1/63)), a third-order Volterra filter to compensate for the signal nonlinearity in the AWG, up-sampling with a root-raised cosine filter with a roll-off factor of 0.02, and digital linear pre-emphasis. As shown in Fig. 2(a), thulium- and erbium-doped fiber amplifiers (TDFA and EDFA) were used for the S, C, and L bands. The wavelength-tunable laser diode (LD) for the main signal was an external cavity laser (ECL). The center wavelength (frequency) of the main signal was swept, assuming a WDM grid of 200 GHz, as follows: 1467.772–1526.050 nm (196.450–204.250 THz), 1530.725–1565.905 nm (191.450–195.850 THz), and 1570.828–1618.313 nm (185.250–190.850 THz) in the S, C, and L bands, respectively. The total number of WDM channels was 92, with 40, 23, and 29 in the S, C, and L bands, respectively. The total WDM bandwidth was 18.4 THz, with 8.0, 4.6, and 5.8 THz in the S, C, and L bands, respectively. The modulation format of the main signal was 192-GBaud PCS-16QAM with the constellation entropy per two-dimensional (2D) symbol of 3.74 bits and PCS-64QAM with 5.25 bits. As

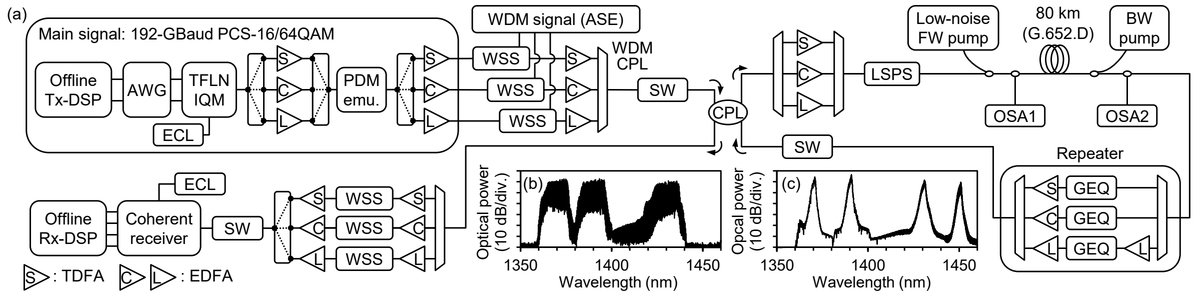

Fig. 2. (a) Experimental setup for 100-Tb/s-class long-haul transmission with 192-GBaud signal in 18.4-THz S+C+L WDM configuration utilizing bidirectional DRA where spectra of proposed low-noise FW pump and BW pump are shown in (b) and (c).

shown in the inset of Fig. 3(a), the 192-GBaud main signal, assuming the 200-GHz WDM grid, was multiplexed with a WDM signal emulated using amplified spontaneous emission (ASE) from TDFA and EDFA, in a flexible-grid wavelength-selective switch (WSS) for each waveband. The frequency amplitude response of the main signal was optically compensated for in the WSS. Finally, an 18.4-THz triple-band WDM output from a WDM coupler (CPL) was fed into a transmission line through an optical switch (SW) and a 3-dB CPL.

A re-circulating loop was used for the transmission line, consisting of optical amplifiers of TDFA for the S band and EDFAs for the C and L bands, a loop-synchronous polarization scrambler (LSPS), our proposed low-noise FW pump unit [10, 16, 17] for DRA, an 80-km low-water peak fiber compliant with ITU-T G.652.D, a backward (BW) pump unit for DRA, an optical repeater with WDM CPLs, a TDFA and EDFAs, and gain equalizers (GEQs) based on flexible-grid WSS, and an optical SW. The WDM launch power, spectral slope, and FW and BW pump powers were calculated on the basis of an optimization scheme that maximizes total throughput [18] using a closed-form GN model [19]. In accordance with the optimization scheme, the fiber input power for each S-, C-, and L-band WDM signal was set to 45.6 mW (16.59 dBm), 18.3 mW (12.62 dBm), and 26.9 mW (14.30 dBm), respectively. The WDM spectral shape was also set to the calculated spectral slope using the GEQ for each waveband in the optical repeater (see Fig. 3(a)). The pump powers for our proposed low-noise FW pumps were set to 201.4, 210.0, and 145.0 mW with the center wavelengths of 1370, 1390, and 1430 nm, respectively. The pump powers in the BW pump unit were set to 306, 455, 185, and 50 mW for conventional coherent pumps with the center wavelengths of 1370, 1390, 1430, and 1450 nm, respectively. Considering safety operations to avoid eye and fire hazards, we limited the total input power for each FW and BW direction to less than 1 W. The spectra of the proposed low-noise FW pump and BW pump are shown in Fig. 2(b) and (c), respectively. The FW pumps were structured with polarization-interleaved narrow longitudinal modes using high-power Fabry-Perot LDs without a fiber Bragg grating, reducing the noise transfer from the pumps to the signal [10, 16, 17].

On the receiver (Rx) side, the S+C+L WDM signal was divided for each waveband signal in a WDM CPL, and the main signal was then demultiplexed from the WDM signal in a WSS. After cutting a surge of the main signal by an optical SW, the main signal was detected in a coherent receiver with four 100-GHz balanced photodetectors and a 256-GSa/s 113-GHz digital storage oscilloscope. An ECL was also used as the optical local oscillator. In the offline Rx-DSP, the received signal was resampled to two samples per symbol and then equalized by using an 8×2 MIMO frequency-domain adaptive equalizer with a pilot-based digital phase-locked loop [20]. After equalization, we measured the normalized generalized mutual information (NGMI) from log-likelihood ratios (LLRs) to estimate the achievable bitrate. We also estimated the code rate after forward error correction decoding with LLRs by using the rate-adaptive coding technique [21] to calculate the net bitrate. The achievable and net bitrates for each wavelength channel were derived from the following equation: $C = [2B\{H_{2D} - (1 - R)\log_2 M\}] \,/\, (1 + P_{OH})$ (1), where B is the symbol rate, H_{2D} is the constellation entropy per 2D symbol, R is the NGMI and code rate, M is the modulation order, and P_{OH} is the pilot overhead.

3. Results and Discussion

Figure 3(a) and (b) show WDM spectra before and after transmission at the first and twenty-fifth span at the monitor point of optical spectrum analyzer 1 (OSA1) and OSA2, respectively. The GEQ for each waveband in the optical repeater worked well to match the optimum spectral shape. Figure 3(c) shows the Raman on-off gain in the re-circulating loop. Thanks to our proposed low-noise FW-pumped DRA technique [10, 16, 17] in conjunction with BW-pumped DRA, we obtained sufficient Raman on-off gain in the S and C bands ranging from 16.7 to 24.7 dB (including FW Raman on-off gain from 1.2 to 4.9 dB) to compensate for the loss of the 80-km G.652.D fiber. Therefore, as shown in Fig. 2(a), the S and C bands did not need to be pre-amplified before optical GEQ in the optical repeater; all-Raman transmission was achieved in the S and C bands. As reported in our previous work [10], this configuration

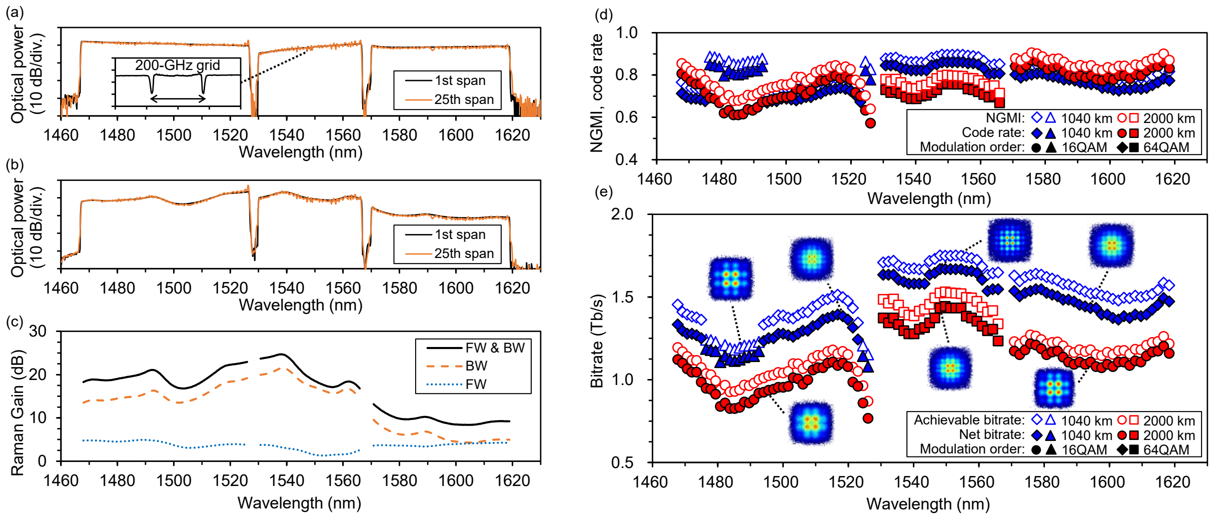

Fig. 3. Experimental results: WDM spectra (a) before and (c) after transmission at first and twenty-fifth span (80 and 2000 km) at monitor points of OSA1 and OSA2, respectively. (c) Raman on-off gain of FW and BW, BW, and FW DRA. FW Raman gain is derived from the difference between FW and BW gain and BW gain. (d) NGMI and code rate and (e) achievable and net bitrates for each wavelength channel after 1040 and 2000 km transmission in 18.4-THz S+C+L-band WDM with 92 channels of 192-GBaud PCS-16/64QAM signals.

enables long-haul 2000-km transmission with beyond 100-Tb/s SMF capacity.

We evaluated the long-haul transmission performance over 1040 and 2000 km in the 18.4-THz WDM configuration with 92 channels of the 192-GBaud signals with PCS-16QAM and PCS-64QAM (see Figs. 3(d) and 3(e)). The NGMI and code rate for each waveband were shown in Fig. 3(d). The achievable and net bitrates for each wavelength channel, shown in Fig. 3(e), were estimated by equation (1) using the NGMI and code rate. The total achievable (and net) bitrate was 138.0 (129.3) Tb/s over 1040 km and 110.6 (102.8) Tb/s over 2000 km. As a result, we obtained the highest average channel rate (average net bitrate per wavelength) of 1.406 Tb/s/λ over 1040 km and 1.117 Tb/s/λ over 2000 km in the 100-Tb/s-class long-haul transmission, as shown in Fig. 1.

4. Conclusion

We have successfully demonstrated high-symbol-rate 192-GBaud signal transmission using 92 channels in an 18.4-THz S+C+L-band WDM configuration over 1040 km with net bitrates of 129.3 Tb/s and over 2000 km with 102.8 Tb/s. Thanks to our low-noise FW-pumped DRA technique with the high-symbol-rate 192-GBaud PCS-16/64QAM signals, we achieved, to the best of our knowledge, the highest average channel rate of 1.406 Tb/s/λ at 1040 km and 1.117 Tb/s/λ at 2000 km for 100-Tb/s-class SMF inline-amplified transmission.

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References

[1] M. Nakamura *et al*., "46.4-Tb/s full C-band 246-km transmission with net >2-Tb/s/λ WDM signals using ...," ECOC2024, Tu4B. 5.
[2] H. Mardoyan *et al*., "First single-carrier transmission at net data rates of 1.6 Tb/s over 9075 km and 2.4 Tb/s ...," ECOC2024, Th2B.
[3] M. Nakamura *et al*., "Net bitrate of 2.52-Tb/s/λ 120-km single-channel transmission and >2-Tb/s/λ 1040-km ...," ECOC2025, Th.03.02.1.
[4] F. Hamaoka *et al*, "Long-haul >100-Tb/s transmission over >1000 km ...," J. Lightw. Technol., 43(4). 1987 (2024).
[5] M. Ionescu *et al*., "91 nm C+L hybrid distributed Raman/erbium-doped fibre amplifier for high capacity ...," ECOC2018, Mo4G.2.
[6] J. X. Cai *et al*., "94.9 Tb/s single mode capacity demonstration over 1,900 km with C+L EDFAs ...," ECOC2018, Mo4G.3.
[7] Y. Yu *et al*., "100.5 Tb/s MLC-CS-256QAM transmission over 600-km single mode fiber with C+L band EDFA," ACP2018.
[8] F. Hamaoka *et al*., "110.7-Tb/s single-mode-fiber transmission over 1040 km with high-symbol-rate 144-GBaud ...," OFC2024, Tu3E.6.
[9] F. Hamaoka *et al*., "Net 107.7-Tb/s triple-band WDM transmission over 1200-km single-mode fiber with forward- ...," OFC2025, Th3C.2.
[10] F. Hamaoka *et al*., "Long-haul 2000-km single-mode fibre transmission with net bitrate of 105.6 Tb/s in S+C+L ...," ECOC2025, Tu.03.05.2.
[11] J. Yang *et al*., "On the feasibility of SCL-band transmission over G.654.E-compliant long-haul fibre links", ECOC2025, Th.02.06.3.
[12] T. Kobayashi *et al*., "C+L+U-band 14.85-THz WDM transmission over 80-km-span G.654.E fiber ...," OFC2024, Th4A.1.
[13] S. Shimizu *et al*., "133-Tbps 1040-km (13×80 km) lumped-amplified transmission over 22 THz in S-to-U-band ...," ECOC2024, Th3B.2.
[14] S. Shimizu *et al*., "18.6-THz long-haul transmission over G.654.E cutoff-shifted fiber across ...," Opt. Express, 33(20), 43096 (2025).
[15] S. Shimizu *et al*., "27-THz ISRS-supported transmission over 1040 km in S+C+L+U and extreme longer-wavelength ...," OFC2025, Th4A.2.
[16] H. Kawakami *et al*., "Multiple beat-noise suppression in polarization-multiplexed pump light ...," J. Lightw. Technol., 41(12). 3892 (2023).
[17] H. Kawakami *et al*., "Forward Pumping with Optimized Longitudinal Modes for Distributed Raman Amplifier ...," OECC2025, TuB2-4.
[18] K. Kimura *et al*., "100-Tb/s-class C+L+U-Band NZ-DSF transmission with ...," J. Lightw. Technol., 43(13). 6313 (2025).
[19] Y. Jiang *et al*., "Closed-Form EGN Model with Comprehensive Raman Support," ECOC2024, W1B.1.
[20] M. Nakamura *et al*., "High information rate of 128-GBaud 1.8-Tb/s and 64-GBaud 1.03-Tb/s signal generation ...," OFC2022, M3H.1.
[21] M. Nakamura *et al*., "Net-bit rate of >562-Gb/s with 32-GBaud probabilistically constellation-shaped 1024QAM ...," ECOC2022, Th3C.1.